A CAUCHY LIMIT FOR A SAWTOOTH SUM

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ABSTRACT. For $S_n(\alpha) = \sum_{k \le n}(1/2 - \{k\alpha\})$, with α uniformly distributed on $(0, 1)$, we prove that $S_n(\alpha)/\log n$ converges in distribution to the centered Cauchy law of scale $1/(2\pi)$. An exact Euclidean-algorithm reciprocity formula reduces the sum to an alternating continued-fraction cost. Its principal part is a Gauss digit marked by a rapidly oscillating torus coordinate, while the carry remainder has uniformly bounded summands. Local cylinder oscillation and a resonance decomposition yield a Poisson limit for the large jumps and a uniform variance bound for the compensated small jumps; a pullback carry construction gives a weak law for the bounded remainder. Consequently the limiting distribution function is $\frac{1}{2} + \pi^{-1}\arctan(2\pi c)$.

1. Introduction and statement of the result

Write $\{x\} = x - \lfloor x \rfloor$ and let logarithms be natural. For $n \in \mathbb{N}_0$, put

$$S_n(\alpha) = \sum_{k=1}^{n} \left(\frac{1}{2} - \{k\alpha\} \right), \qquad 0 < \alpha < 1,$$

with the empty-sum convention $S_0 = 0$. For $n \ge 1$, the question is whether $S_n(\alpha)/\log n$, regarded as a random variable on the Lebesgue probability space $(0, 1)$, has a limiting law. This is Erdős Problem #1002, originating in [5]; see also [2]. Erdős used the opposite sign convention; since $\alpha \mapsto 1 - \alpha$ preserves Lebesgue measure and reverses the sum for irrational α, the two distributional formulations are equivalent. Kesten's two-parameter result [8] provides the closely related averaged-origin case; Borda [3] computed the normalizing constant in that law for the sawtooth observable. This is an annealed spatial law, averaging both the rotation number and the starting point. Dolgopyat and Sarig [4] proved an annealed temporal Cauchy law in which the rotation number and the time are randomized. Neither ensemble is the one considered here: the time n and the starting point are fixed, and only α is averaged. A related Cauchy limit phenomenon occurs in Vardi's theorem for Dedekind sums [10], although the observable and averaging ensemble are different from those considered here. The answer to this fixed-origin, fixed-time problem is as follows.

Theorem 1.1. *As $n \to \infty$,*

$$\frac{S_n(\alpha)}{\log n} \implies \mathrm{Cauchy}\left(0, \frac{1}{2\pi}\right).$$

Equivalently, for every $c \in \mathbb{R}$,

$$\lim_{n\to\infty} \mathrm{Leb}\left\{ \alpha \in (0,1) : \frac{S_n(\alpha)}{\log n} \le c \right\} = \frac{1}{2} + \frac{1}{\pi}\arctan(2\pi c).$$

Here $\mathrm{Cauchy}(0, \gamma)$ denotes the centered Cauchy law with characteristic function $t \mapsto \exp(-\gamma|t|)$.

The proof has four parts. First an exact reciprocity identity expresses S_n as an alternating sum along the Gauss map. Second, continued-fraction coordinates separate each summand into a heavy-tailed marked digit and a bounded carry remainder. Third, the marked digits satisfy a Poisson limit; the main point is that the initial measure lies on the oscillating graph $\alpha \mapsto (\alpha, \{n\alpha\})$, so ordinary Gauss mixing is insufficient. We instead integrate Fourier modes on complete continued-fraction cylinders, before and after their unique resonance scale. Finally, the

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carry is realized as a measurable graph over the natural extension of a Gauss–torus cocycle; a reset argument and a two-block correlation estimate give a weak law for the bounded remainder.

Throughout, constants denoted by C, c may change from line to line. A bound $O_A(L^{-A})$ is uniform in the time indices satisfying the displayed hypotheses. We put $L = \log n$ whenever n is fixed.

2. Exact renormalization and continued-fraction coordinates

Let $T : (0,1) \to (0,1)$ be the Gauss map,

$$Tx = \left\{ \frac{1}{x} \right\}, \qquad a_1(x) = \left\lfloor \frac{1}{x} \right\rfloor.$$

Rational points will always be ignored, since they form a null set.

Proposition 2.1 (Reciprocity)**.** *For irrational $x \in (0,1)$ and an integer $N \geq 1$, set*

$$m = \lfloor Nx \rfloor, \qquad u = \{Nx\}.$$

Then

$$S_N(x) = -S_m(Tx) + \Phi(x,u), \qquad \Phi(x,u) = \frac{u(1-u)}{2x} - \frac{u}{2}. \tag{1}$$

Proof. Since x is irrational,

$$S_N(x) = \frac{N}{2} - \frac{xN(N+1)}{2} + \sum_{k=1}^{N} \lfloor kx \rfloor.$$

Counting lattice points below $j = kx$ in the other order gives

$$\sum_{k=1}^{N} \lfloor kx \rfloor = Nm - \sum_{j=1}^{m} \left\lfloor \frac{j}{x} \right\rfloor.$$

If $a = \lfloor 1/x \rfloor$, then $1/x = a + Tx$, and hence

$$\sum_{j=1}^{m} \left\lfloor \frac{j}{x} \right\rfloor = \frac{am(m+1)}{2} + \sum_{j=1}^{m} \lfloor jTx \rfloor.$$

Substitution and the identity $m = Nx - u$ reduce the remaining polynomial expression to $u(1-u)/(2x) - u/2$. ☐

Starting with $N_0 = n$, $x_0 = \alpha$, define

$$N_{j+1} = \lfloor N_j x_j \rfloor, \qquad x_{j+1} = Tx_j, \qquad u_j = \{N_j x_j\}. \tag{2}$$

Let $\tau_n = \min\{j : N_j = 0\}$. Iterating Proposition 2.1 gives the exact identity

$$S_n(\alpha) = \sum_{j=0}^{\tau_n - 1} (-1)^j \Phi(x_j, u_j). \tag{3}$$

Write

$$\alpha = [0; a_1, a_2, \ldots], \qquad x_j = T^j \alpha, \qquad a_{j+1} = a_1(x_j),$$

and let p_j/q_j be the convergents, normalized by

$$p_{-1} = 1, \quad p_0 = 0, \qquad q_{-1} = 0, \quad q_0 = 1.$$

The standard identities

$$\beta_j := x_0 x_1 \cdots x_j = (-1)^j (q_j \alpha - p_j) = \frac{1}{q_{j+1} + q_j x_{j+1}} \tag{4}$$

will be used repeatedly; set $\beta_{-1} = 1$.

Define the continuous heights and their errors by

$$C_j = n\beta_{j-1}, \qquad E_j = C_j - N_j.$$

Then $C_{j+1} = x_j C_j$, and (2) yields

$$E_{j+1} = x_j E_j + u_j, \qquad E_0 = 0. \tag{5}$$

For every $r, \ell \geq 0$,

$$x_r x_{r+1} \cdots x_{r+\ell-1} \leq F_{\ell+1}^{-1}, \tag{6}$$

where $F_1 = F_2 = 1$. Indeed, the reciprocal of the product is a continuant with all digits at least one. Iterating (5) therefore gives

$$0 \leq E_j \leq E_* := \sum_{\ell \geq 1} F_\ell^{-1} < \infty \tag{7}$$

uniformly in j, n, α.

Put

$$\theta_j = \{n\beta_j\} = \{(-1)^j n q_j \alpha\}, \qquad \theta_{-1} = 0. \tag{8}$$

From $q_{j+1} = a_{j+1} q_j + q_{j-1}$ we obtain

$$\theta_{j+1} = \{\theta_{j-1} - a_{j+1}\theta_j\}. \tag{9}$$

Since $\{E_j\} = \theta_{j-1}$, write

$$E_j = d_j + \theta_{j-1}, \qquad d_j = \lfloor E_j \rfloor.$$

The equality $C_{j+1} = N_{j+1} + u_j + x_j E_j$ implies

$$u_j = \{\theta_j - x_j(d_j + \theta_{j-1})\}. \tag{10}$$

Let

$$W(t) = \frac{\{t\}(1 - \{t\})}{2}, \qquad t \in \mathbb{R}/\mathbb{Z}.$$

This function is 1/2-Lipschitz and $0 \leq W \leq 1/8$.

Proposition 2.2 (Principal term)**.** *There is an absolute constant C_0 such that, for every j,*

$$\Phi(x_j, u_j) = a_{j+1} W(\theta_j) + B_j, \qquad |B_j| \leq C_0. \tag{11}$$

Proof. By (10) and (7),

$$\mathrm{dist}_{\mathbb{T}}(u_j, \theta_j) \leq x_j E_*.$$

Since $x_j^{-1} = a_{j+1} + x_{j+1}$,

$$\Phi(x_j, u_j) - a_{j+1} W(\theta_j) = a_{j+1}\big(W(u_j) - W(\theta_j)\big) + x_{j+1} W(u_j) - \frac{u_j}{2}.$$

The first term has absolute value at most $a_{j+1} x_j E_*/2 \leq E_*/2$, and the other two are bounded. □

3. Gauss estimates and local oscillation

The Gauss invariant probability measure is

$$d\nu(x) = h(x)\, dx, \qquad h(x) = \frac{1}{\log 2\,(1+x)}. \tag{12}$$

The constant

$$\lambda = \int_0^1 -\log x \, d\nu(x) = \frac{\pi^2}{12 \log 2} \tag{13}$$

is the Lévy constant. Since $\log |T'(x)| = -2\log x$, the Lyapunov exponent of the Gauss map is 2λ.

For a word $w = (a_1, \ldots, a_r)$ let I_w be the corresponding continued-fraction cylinder. If p_r/q_r is its last convergent, the inverse branch is

$$h_w(y) = \frac{p_r + p_{r-1} y}{q_r + q_{r-1} y}, \qquad |h_w'(y)| = (q_r + q_{r-1} y)^{-2},$$

and

$$|I_w| = \frac{1}{q_r(q_r + q_{r-1})}. \tag{14}$$

Conditionally on I_w, the density of $T^r\alpha = y$ is

$$\rho_w(y) = \frac{q_r(q_r + q_{r-1})}{(q_r + q_{r-1}y)^2}.$$

Thus $\sup_w \|\rho_w\|_{BV} < \infty$.

We use the following standard consequences of the Lasota–Yorke inequality for the Gauss transfer operator; see, for example, [1]. They are recorded with the uniformity needed below.

Lemma 3.1 (Gauss estimates)**.** *There are $C, c > 0$ and $0 < \rho < 1$ such that:*

(1) *for every integer $r \geq 0$ and $f \in BV(0,1)$,*

$$\|\mathcal{L}^r f - h\int f\|_1 \leq C\rho^r \|f\|_{BV};$$

(2) *for an integer $A \geq 1$, conditionally on any past cylinder,*

$$\mathbb{P}(a_{j+1} \geq A \mid a_1, \dots, a_j) \leq C/A;$$

in particular, for distinct $j_1 < \cdots < j_s$ and positive integers $A_1, \dots, A_s$,

$$\mathbb{P}(a_{j_i+1} \geq A_i,\ 1 \leq i \leq s) \leq C^s \prod_{i=1}^{s} A_i^{-1}; \tag{15}$$

(3) *uniformly for $r \geq 1$ and $v > 0$,*

$$\mathbb{P}(|\log q_r - \lambda r| > v) \leq C \exp\{-c \min(v^2/r, v)\}. \tag{16}$$

Proof. For completeness, the transfer operator is

$$\mathcal{L}f(x) = \sum_{a\geq 1}(a+x)^{-2} f\big((a+x)^{-1}\big).$$

On two-step inverse branches the contraction is at most $1/2$; the sums of the inverse Jacobians and of their variations converge. The branchwise chain rule gives

$$\mathrm{Var}(\mathcal{L}^2 f) \leq \rho_0 \mathrm{Var}(f) + C\|f\|_1, \qquad \rho_0 < 1.$$

Helly compactness and exactness yield (i). The conditional densities ρ_w have uniformly bounded variation, and $\{a_1 \geq A\} = (0, 1/A]$ up to endpoints; this proves (ii), and iteration proves (15). Finally, the perturbed operators

$$\mathcal{L}_t f(x) = \sum_{a\geq 1}(a+x)^{-2+t} f\big((a+x)^{-1}\big)$$

form an analytic family on BV for $|t| < 1/2$. Indeed, on every smaller complex disc the branch series and all its t-derivatives converge in operator norm, since their bounds reduce to convergent series of the form $\sum_{a\geq 1} a^{-2+t_0}(1 + (\log a)^k)$. Analytic perturbation of the simple leading eigenvalue [7, Chap. VII] therefore gives, after decreasing $t_0 > 0$ if necessary,

$$\mathcal{L}_t = \kappa(t)\Pi_t + N_t, \qquad \Pi_t N_t = N_t \Pi_t = 0,$$

$$\sup_{|t|\leq t_0} \|\Pi_t\|_{BV\to BV} < \infty, \qquad \sup_{|t|\leq t_0} \|N_t^r\|_{BV\to L^1} \leq C\rho_1^r \quad (r \geq 0).$$

where these bounds are for real t and $0 < \rho_1 < \min_{|t|\leq t_0} \kappa(t)$. Moreover,

$$\kappa'(0) = \int_0^1 -\log x \, d\nu(x) = \lambda, \qquad \mathbb{E}\exp\left(t\sum_{i<r} -\log T^i\alpha\right) = \int_0^1 \mathcal{L}_t^r \mathbf{1}\, dx.$$

Consequently,

$$\log \kappa(t) = \lambda t + O(t^2), \qquad \mathbb{E}\exp\left(t\sum_{i<r} -\log T^i\alpha\right) \leq C\kappa(t)^r$$

for $|t|$ sufficiently small; the same estimate holds for negative t. For $0 < v \le r$, Chernoff's inequality with $|t| \asymp v/r$ gives $C\exp(-cv^2/r)$, while for $v > r$ a fixed sufficiently small $|t|$ gives $C\exp(-cv)$. Thus the asserted Bernstein bound holds for $\sum_{i<r} -\log T^i\alpha$. Since

$$\left|\log q_r - \sum_{i<r} -\log T^i\alpha\right| \le \log 2,$$

we obtain (iii). $\square$

Lemma 3.2 (Conditional multi-block mixing). *Let $s, M \ge 1$ and $d \ge 0$ be integers, let I_w be a cylinder of depth d, and let $t_1, \dots, t_s$ be integers satisfying $d + M \le t_1 < \cdots < t_s$ and $t_{i+1} - t_i \ge M$. If $g_1, \dots, g_s \in BV(0,1)$, then*

$$\left|\mathbb{E}\left(\prod_{i=1}^s g_i(T^{t_i}\alpha)\,\middle|\, I_w\right) - \prod_{i=1}^s \int g_i\,d\nu\right| \le C_s\rho^M \prod_{i=1}^s \|g_i\|_{BV}. \tag{17}$$

The constants are uniform in the cylinder I_w.

Proof. Conditioned on I_w, the density of $T^d\alpha$ is ρ_w, whose BV norm is uniformly bounded by the displayed formula for ρ_w above. More precisely, change of variables by the inverse branch of I_w gives the exact cylinder-transfer identity

$$\mathbb{E}\left(\prod_{i=1}^s g_i(T^{t_i}\alpha)\,\middle|\, I_w\right) = \int_0^1 \prod_{i=1}^s g_i(T^{t_i-d}y)\rho_w(y)\,dy.$$

Apply Lemma 3.1(i) across the first gap, multiply by the next BV observable, and iterate. The product inequality $\|fg\|_{BV} \le C\|f\|_{BV}\|g\|_{BV}$ and the fixed number s of factors give (17). $\square$

Fix henceforth

$$L = \log n, \qquad H = L^{3/4}, \qquad m_n = \lfloor L/\lambda \rfloor, \tag{18}$$

and define the bulk set

$$\mathcal{J}_n = \{j \in \mathbb{Z} : 200H \le j \le m_n - 200H\}. \tag{19}$$

By (16), for every fixed $\delta > 0$ and every integer $0 \le t \le 2m_n$,

$$\mathbb{P}(q_t \notin [\mathrm{e}^{\lambda t - \delta H}, \mathrm{e}^{\lambda t + \delta H}]) \le C_\delta \mathrm{e}^{-c_\delta L^{1/2}}. \tag{20}$$

We always use (20) as a union of complete depth-t cylinders. We never insert a global good-event indicator inside an oscillatory integral.

The first ingredient controls local cancellation of the two continuants in a torus character.

Lemma 3.3 (Shrinking anti-concentration). *For every $(r,s) \in \mathbb{Z}^2 \setminus \{(0,0)\}$, every $j \ge 1$, and every $\eta > 0$,*

$$\mathbb{P}(|sq_j - rq_{j-1}| < \eta q_j) \le 4\eta + 4F_{j+1}^{-2} \le C\big(\eta + \mathrm{e}^{-2(\log\varphi)j}\big), \qquad \varphi = (1+\sqrt{5})/2. \tag{21}$$

Proof. Put $Y_j = q_{j-1}/q_j = [0; a_j, \dots, a_1]$. For a word $w = (a_1, \dots, a_j)$, let $w^{\mathrm{rev}} = (a_j, \dots, a_1)$. Symmetry of the continuant gives

$$q_j(w^{\mathrm{rev}}) = q_j(w), \qquad p_j(w^{\mathrm{rev}}) = q_{j-1}(w), \qquad q_{j-1}(w^{\mathrm{rev}}) = p_j(w).$$

Consequently $Y_j(w) = p_j(w^{\mathrm{rev}})/q_j(w^{\mathrm{rev}})$, while (14) gives the uniform distortion comparison

$$\frac{|I_w|}{|I_{w^{\mathrm{rev}}}|} = \frac{q_j(w) + p_j(w)}{q_j(w) + q_{j-1}(w)} \le 2.$$

The rational $p_j(w^{\mathrm{rev}})/q_j(w^{\mathrm{rev}})$ is an endpoint of $I_{w^{\mathrm{rev}}}$, and $q_j \ge F_{j+1}$. Thus, if $I \subset (0,1)$ is an interval and $\delta_j = F_{j+1}^{-2}$, every reversed cylinder contributing to $\{Y_j \in I\}$ lies in the δ_j-neighborhood $I^{(\delta_j)}$. Since word reversal is a bijection and the depth-j cylinders are disjoint up to endpoints,

$$\mathbb{P}(Y_j \in I) \le 2\sum_{w : Y_j(w) \in I} |I_{w^{\mathrm{rev}}}| \le 2|I^{(\delta_j)}| \le 2|I| + 4\delta_j.$$

If $r \neq 0$, the event in (21) is $|s - rY_j| < \eta$ and its possible Y_j-values lie in an interval of length at most $2\eta/|r| \leq 2\eta$. If $r = 0$, then $s \neq 0$; the event is empty when $\eta \leq |s|$, while otherwise its probability is $1 \leq \eta$. This proves the first bound uniformly in (r, s). The second follows from $F_{j+1}^{-2} \ll \varphi^{-2j}$. □

The second ingredient is a complete-cylinder oscillatory estimate. Its formulation makes the measurability needed for integration by parts explicit.

Lemma 3.4 (Descendant-cylinder estimate). *Let $\varepsilon > 0$, and let $\{I_w\}$ be cylinders of a fixed depth d. On I_w let $Q_w \in \mathbb{Z} \setminus \{0\}$ be fixed. Let $k > d$ and let A be constant on every depth-k cylinder, with $|A| \leq 1$. Suppose that, below I_w, the support of A is a union of depth-k descendants satisfying $q_k \leq R_w$, where $R_w > 0$ and*

$$R_w^2 \leq \varepsilon n |Q_w|.$$

Then

$$\left| \sum_w \int_{I_w} \mathrm{e}^{2\pi i n Q_w \alpha} A(\alpha)\, d\alpha \right| \leq C\varepsilon. \tag{22}$$

Conversely, let $Q \in \mathbb{Z} \setminus \{0\}$. If a depth-$k$ cylinder J satisfies $q_k^2 \geq \varepsilon^{-1} n|Q|$, then

$$\sup_{\alpha,\alpha' \in J} |nQ(\alpha - \alpha')| \leq C\varepsilon. \tag{23}$$

Proof. Let $q(w)$ and $q_-(w)$ be the last two denominators of the prefix w. If (ℓ, ℓ') is the denominator pair of a continuation, the final denominator is $q_k = q(w)\ell + q_-(w)\ell' \geq q(w)\ell$. The number of descendants with $q_k \leq R_w$ is therefore $O(R_w^2/q(w)^2)$. On each complete descendant cylinder J,

$$\left| \int_J \mathrm{e}^{2\pi i n Q_w \alpha}\, d\alpha \right| \leq (\pi n |Q_w|)^{-1}.$$

Thus the contribution below w is at most $C\varepsilon/q(w)^2$. By (14), $q(w)^{-2} \leq 2|I_w|$, and summation proves (22). The cylinder length bound $|J| \leq q_k^{-2}$ gives (23). □

4. Marked factorization and resonances

For $D > 0$ let $\mathcal{P}_D(L)$ consist of functions

$$F(a, \theta) = \sum_{|v| \leq L^D} c(a, v) \mathrm{e}^{2\pi i v \theta}, \qquad c(a, v) = 0 \quad (a > L^D), \tag{24}$$

such that $\sum_{a,v} |c(a, v)| \leq L^D$. For times $j_1 < \cdots < j_r$ in $\mathcal{J}_n$, call the tuple good if

$$j_{\ell+1} - j_\ell \geq 200H \tag{25}$$

and, for every $i < \ell$,

$$\left| j_\ell - \frac{m_n + j_i}{2} \right| \geq 200H. \tag{26}$$

The number of non-good r-tuples is $O_r(L^{r-1}H)$.

Proposition 4.1 (Marked factorization). *Fix an integer $r \geq 1$ and $D, A > 0$. If $(j_1, \ldots, j_r)$ is good and $F_1, \ldots, F_r \in \mathcal{P}_D(L)$, then*

$$\begin{aligned} &\int_0^1 \prod_{\ell=1}^r F_\ell(a_{j_\ell+1}, \theta_{j_\ell})\, d\alpha \\ &\qquad = \prod_{\ell=1}^r \int_0^1 \int_0^1 F_\ell(a_1(x), \theta)\, d\theta\, d\nu(x) + O_{r,D,A}(L^{-A}). \end{aligned} \tag{27}$$

Proof. Expand both in the digit values and in the Fourier modes $v_1, \dots, v_r$, and pull out the scalar coefficient of each digit–Fourier term. The remaining amplitude is a product of cylinder indicators and hence has modulus at most one; the total ℓ^1-mass of the extracted coefficients is at most L^{Dr}. If every $v_\ell = 0$, the integrand is a product of digit functions. For each such factor,

$$g_\ell(x) = \sum_a c_\ell(a,0)\mathbf{1}_{\{a_1(x)=a\}}, \qquad \mathrm{Var}(g_\ell) \le 2\sum_a |c_\ell(a,0)| \le 2L^D.$$

This variation estimate comes from the coefficient ℓ^1 bound in (24), and in particular $\|g_\ell\|_{BV} \ll L^D$; merely counting its jumps would not give the stated bound. Repeated use of Lemma 3.2 and (25) factors its mean with error $L^{O_{r,D}(1)}\rho^{200H} = O_A(L^{-A})$.

Suppose now that $v_s \neq 0$ and $v_\ell = 0$ for $\ell > s$. By (8), the phase is

$$\mathrm{e}^{2\pi i n Q\alpha}, \qquad Q = \sum_{\ell=1}^{s} v_\ell(-1)^{j_\ell} q_{j_\ell}.$$

The continuants grow at least at the Fibonacci rate. The separation condition and $|v_\ell| \le L^D$ therefore give, deterministically,

$$\tfrac{1}{2}q_{j_s} \le |Q| \le 2L^D q_{j_s} \tag{28}$$

for all sufficiently large n.

The nominal resonance time is

$$R_s = \frac{m_n + j_s}{2}.$$

Put $k_\pm = \lfloor R_s \pm 40H \rfloor$. We now make only local complete-cylinder cuts. At depth j_s retain the cylinder union on which

$$\mathrm{e}^{\lambda j_s - \delta H} \le q_{j_s} \le \mathrm{e}^{\lambda j_s + \delta H}.$$

At depths k_- and k_+ retain, respectively,

$$q_{k_-} \le \mathrm{e}^{\lambda k_- + \delta H}, \qquad q_{k_+} \ge \mathrm{e}^{\lambda k_+ - \delta H},$$

where $\delta > 0$ is sufficiently small. By (20), the total discarded mass is $O(\mathrm{e}^{-cL^{1/2}})$. On every retained prefix and descendant,

$$q_{k_-}^2 \le \mathrm{e}^{-cH} n|Q|, \qquad q_{k_+}^2 \ge \mathrm{e}^{cH} n|Q|. \tag{29}$$

The factors after s are zero torus modes and hence digit functions. By (26), each lies either at least $100H$ before k_- or at least $100H$ after k_+. For a fixed digit–Fourier term, write A_- for the product of all pre-resonance factors together with the retained depth-k_- cutoff, write A_+ for the product of the post-resonance digit factors, denote those digit functions by g_ℓ, and put

$$P_+ = \prod_{\substack{\ell > s \\ j_\ell \ge k_+ + 100H}} \int g_\ell \, d\nu.$$

Partition into complete depth-k_+ cylinders. On every retained such cylinder J, phase freezing and conditional mixing give

$$\int_J \mathrm{e}^{2\pi i n Q\alpha} A_-(\alpha)A_+(\alpha)\, d\alpha = P_+ \int_J \mathrm{e}^{2\pi i n Q\alpha} A_-(\alpha)\, d\alpha + O\Big(|J| L^{O_{r,D}(1)}(\mathrm{e}^{-cH} + \rho^{cH})\Big).$$

Here the phase-freezing error is (23), and the mixing error is summed by cylinder mass before the extracted coefficient ℓ^1-mass is restored.

Extend this stationary-mean replacement to the discarded depth-k_+ cylinders and restore them. Their total mass is $O(\mathrm{e}^{-cL^{1/2}})$, so this costs at most $L^{O_{r,D}(1)}\mathrm{e}^{-cL^{1/2}}$ and removes every future cutoff. The remaining amplitude A_- is constant on complete depth-k_- descendants, its support still satisfies the q_{k_-} cutoff, and Q is fixed on each depth-(j_s+1) prefix. The first inequality in (29) and Lemma 3.4 therefore give a contribution $O(\mathrm{e}^{-cH})$. Restoring the other discarded cylinder unions, and summing all polynomially many Fourier terms, gives the uniform bound

$$\delta_n := L^{O_{r,D}(1)}\big(\mathrm{e}^{-cL^{1/2}} + \mathrm{e}^{-cH} + \rho^{cH}\big) = O_A(L^{-A}). \tag{30}$$

Thus every nonzero Fourier term vanishes to the required order, while the zero terms give the product on the right of (27). □

We also need fixed-radius blocks. The recurrence (9) implies by induction that every torus character in the full torus window can be reduced, on its digit cylinder, to a character of the two central coordinates. More precisely, fix $R \geq 0$, an index $j \geq R+1$ for the one-sided process, and integers $c_{-R-1}, \ldots, c_R$, and put

$$w_{j,R} = (a_{j-R+1}, \ldots, a_{j+R}),$$

with the word understood as empty when $R = 0$. There are integers $A_{w_{j,R}}, B_{w_{j,R}}$, depending only on this word, such that

$$\sum_{t=-R-1}^{R} c_t\theta_{j+t} \equiv A_{w_{j,R}}\theta_j + B_{w_{j,R}}\theta_{j-1} \pmod 1. \tag{31}$$

Indeed, the forward substitutions up to θ_{j+R} use precisely $a_{j+1}, \ldots, a_{j+R}$, while the backward substitutions down to θ_{j-R-1} use precisely $a_j, \ldots, a_{j-R+1}$. On each fixed finite collection of these words, A_w, B_w range over a finite set. In view of (8), the resulting phase is $n(-1)^j(A_wq_j - B_wq_{j-1})\alpha$.

Let R, K be fixed and call a cylinder–torus monomial a function of the form

$$U_j(\alpha) = u(a_{j-R+1}, \ldots, a_{j+R})\mathrm{e}^{2\pi i(r\theta_{j-1}+s\theta_j)}, \tag{32}$$

where u is supported on finitely many words and $|r| + |s| \leq K$. Its frequency is

$$(-1)^j Q_j(r, s), \qquad Q_j(r, s) = sq_j - rq_{j-1}. \tag{33}$$

Proposition 4.2 (Two-block factorization). *For two fixed finite linear combinations U, V of monomials* (32)*, all but $O_{U,V}(LH)$ pairs $j < k$ in $\mathcal{J}_n$ satisfy*

$$\left|\int_0^1 U_jV_k\, d\alpha - \int U\, d\widehat{\mu}_0 \int V\, d\widehat{\mu}_0\right| \leq C_{U,V}\big(\mathrm{e}^{-c_{\mathrm{ld}}L^{1/2}} + \mathrm{e}^{-c_{\mathrm{osc}}H} + \rho^{c_{\mathrm{mix}}H}\big), \tag{34}$$

where $c_{\mathrm{ld}}, c_{\mathrm{osc}}, c_{\mathrm{mix}} > 0$ may depend on U, V, and $\widehat{\mu}_0$ is the stationary Gauss natural-extension measure times Haar measure on $\mathbb{T}^2$.

Proof. It suffices to treat monomials; let R be the maximum of their two radii. If both torus modes vanish, ordinary two-sided Gauss mixing proves (34) unless the blocks overlap or are separated by only $O(H)$ positions; these exceptional pairs number $O(LH)$.

Choose $\kappa, \delta > 0$ so that

$$\kappa + 3\delta < 80\lambda, \qquad \gamma := 80\lambda - \kappa - 3\delta > 0.$$

Here κ is used only for anti-concentration and δ only for the denominator windows. Put $c_{\mathrm{anti}} = 2\log\varphi$, as supplied uniformly by Lemma 3.3, and write $c_{\mathrm{ld}} > 0$ for the constant supplied by (20) for this δ.

Assume first that the later mode (r_2, s_2) is nonzero. Apply Lemma 3.3 with $\eta = \mathrm{e}^{-\kappa H}$. Since $k \geq 200H$, outside a complete depth-k cylinder union of mass

$$O(\mathrm{e}^{-\kappa H} + \mathrm{e}^{-200c_{\mathrm{anti}}H}) = O(\mathrm{e}^{-c_{\mathrm{ac}}H}), \qquad c_{\mathrm{ac}} := \min(\kappa, 200c_{\mathrm{anti}}),$$

one has $|Q_k(r_2, s_2)| \geq \mathrm{e}^{-\kappa H}q_k$. If $k-j \geq C_{U,V,\kappa}H$, Fibonacci growth makes the earlier frequency at most half this quantity, and hence

$$|Q| := |(-1)^jQ_j(r_1, s_1) + (-1)^kQ_k(r_2, s_2)| \geq \tfrac{1}{2}\mathrm{e}^{-\kappa H}q_k.$$

The excluded separations again account for only $O_{U,V}(LH)$ pairs. On each depth-$(k+R)$ cylinder, the integer Q and the complete amplitude are fixed. Put

$$t_- := \left\lfloor \frac{m_n+k}{2} - 40H \right\rfloor.$$

Retain the complete depth-k cylinders on which $q_k \geq \mathrm{e}^{\lambda k-\delta H}$ and, below each such cylinder, the complete depth-t_- descendants on which $q_{t_-} \leq \mathrm{e}^{\lambda t_- + \delta H}$. The discarded union has mass $O(\mathrm{e}^{-c_{\mathrm{ld}}L^{1/2}})$. On every retained descendant,

$$\begin{aligned}\log \frac{q_{t_-}^2}{n|Q|} &\leq 2\lambda t_- + 2\delta H - L - \lambda k + (\kappa+\delta)H + O(1) \\ &\leq -(80\lambda - \kappa - 3\delta)H + O(1) = -\gamma H + O(1).\end{aligned}$$

Thus, for all sufficiently large n, $q_{t_-}^2 \leq \mathrm{e}^{-\gamma H/2} n|Q|$. Lemma 3.4, with $\varepsilon = \mathrm{e}^{-\gamma H/2}$, prefix depth $k+R$, and descendant depth t_-, bounds the retained integral by $O(\mathrm{e}^{-\gamma H/2})$. The summation is by complete prefix-cylinder mass, so no cylinder-count factor is lost. The stationary product is zero because the later Haar character is nontrivial.

It remains to suppose that the earlier mode is nonzero and the later mode is zero. Write $Q_j = Q_j(r_1, s_1)$ and

$$t_0 := \frac{m_n + j}{2}, \qquad t_- := \lfloor t_0 - 40H \rfloor, \qquad t_+ := \lfloor t_0 + 40H \rfloor.$$

Apply Lemma 3.3 with the same $\eta = \mathrm{e}^{-\kappa H}$. Outside a complete depth-j cylinder union of mass $O(\mathrm{e}^{-c_{\mathrm{ac}}H})$, one has $|Q_j| \geq \mathrm{e}^{-\kappa H} q_j$. Declare $|k - t_0| \leq 100H$ exceptional. This gives $O(H)$ values of k per j, hence $O(LH)$ pairs in total.

Retain throughout the complete depth-j cylinder union on which

$$\mathrm{e}^{\lambda j - \delta H} \leq q_j \leq \mathrm{e}^{\lambda j + \delta H}.$$

Its complement has mass $O(\mathrm{e}^{-c_{\mathrm{ld}}L^{1/2}})$, and on the retained cylinders

$$\mathrm{e}^{-\kappa H} q_j \leq |Q_j| \leq C_{r_1,s_1} q_j.$$

If $k < t_0 - 100H$, the whole two-block amplitude is measurable before depth t_-. Refine the retained depth-j cylinders into complete depth-$(k+R)$ prefixes, and below them keep the complete depth-t_- descendants with $q_{t_-} \leq \mathrm{e}^{\lambda t_- + \delta H}$. Their complement has mass $O(\mathrm{e}^{-c_{\mathrm{ld}}L^{1/2}})$, while the same calculation as above gives

$$\log \frac{q_{t_-}^2}{n|Q_j|} \leq -\gamma H + O(1).$$

Hence Lemma 3.4, with $\varepsilon = \mathrm{e}^{-\gamma H/2}$, prefix depth $k+R$, and descendant depth t_-, removes the phase with error $O(\mathrm{e}^{-\gamma H/2})$.

If $k > t_0 + 100H$, also retain the complete depth-t_+ cylinders satisfying $q_{t_+} \geq \mathrm{e}^{\lambda t_+ - \delta H}$. Their complement again has mass $O(\mathrm{e}^{-c_{\mathrm{ld}}L^{1/2}})$. Using the upper bound $|Q_j| \leq C_{r_1,s_1} q_j$ gives

$$\log \frac{q_{t_+}^2}{n|Q_j|} \geq (80\lambda - 3\delta)H + O(1).$$

Put $\gamma_+ := 80\lambda - 3\delta > 0$. For all sufficiently large n, $q_{t_+}^2 \geq \mathrm{e}^{\gamma_+ H/2} n|Q_j|$, so (23), with $\varepsilon = \mathrm{e}^{-\gamma_+ H/2}$, makes the earlier phase constant up to $O(\mathrm{e}^{-\gamma_+ H/2})$ on each retained depth-t_+ cylinder. Since the radii of U, V are fixed, the later zero-mode block starts at least $c_{\mathrm{mix}}H$ after t_+ for some $c_{\mathrm{mix}} > 0$ and all large n. Conditional Gauss mixing, applied on each cylinder and summed by cylinder mass, therefore replaces that block by its stationary mean with error $O(\rho^{c_{\mathrm{mix}}H})$. Extend this replacement to the discarded depth-t_+ cylinders and restore them, at cost $O(\mathrm{e}^{-c_{\mathrm{ld}}L^{1/2}})$; no future cutoff indicator then remains in the earlier integral. Finally retain the depth-t_- descendants as above and refine the remaining earlier amplitude to complete depth-$(j+R)$ prefixes. Apply Lemma 3.4, with prefix depth $j+R$ and descendant depth t_-, to the remaining earlier phase.

The stationary product is zero because the earlier Haar character is nontrivial. Taking

$$c_{\mathrm{osc}} := \min(c_{\mathrm{ac}}, \gamma/2, \gamma_+/2)$$

and combining the complete-cylinder errors proves (34). In particular, the denominator-deviation contribution is kept separate from the anti-concentration and oscillatory exponents. □

5. The principal Poisson and stable limits

Let

$$Z_{n,j} = a_{j+1} W(\theta_j).$$

If A has the stationary Gauss digit law and Θ is independent and uniform on $\mathbb{T}$, then

$$\mathbb{P}(AW(\Theta) > t) \sim \frac{c_0}{t}, \qquad c_0 = \frac{1}{12\log 2}. \tag{35}$$

Indeed,

$$\nu(A \geq m) = \frac{1}{\log 2}\log\left(1 + \frac{1}{m}\right) \sim \frac{1}{m\log 2},$$

and dominated convergence gives

$$\lim_{t\to\infty} t\mathbb{P}(AW(\Theta) > t) = \frac{1}{\log 2}\int_0^1 W(\theta)\, d\theta = \frac{1}{12\log 2}.$$

Define the marked point process

$$\Xi_n := \sum_{j\in\mathcal{J}_n} \delta_{(-1)^j Z_{n,j}/L}. \tag{36}$$

Proposition 5.1 (Poisson point process)**.** *In the space of Radon point measures on $\mathbb{R}\setminus\{0\}$ equipped with the vague topology,*

$$\Xi_n \implies \mathrm{PRM}(\Lambda), \qquad d\Lambda(x) = \frac{1}{2\pi^2}|x|^{-2}\, dx. \tag{37}$$

Proof. We verify factorial moment measures against $\varphi_1, \ldots, \varphi_r \in C_c^1(\mathbb{R}\setminus\{0\})$. Fix an auxiliary exponent $A_0 > 0$, to be chosen sufficiently large. Writing $X_{n,j} = (-1)^j Z_{n,j}/L$, the relevant ordered factorial sum is

$$\sum_{\substack{(j_1,\ldots,j_r)\in\mathcal{J}_n^r \\ j_1,\ldots,j_r \text{ pairwise distinct}}} \mathbb{E}\prod_{\ell=1}^r \varphi_\ell(X_{n,j_\ell}). \tag{38}$$

Let $c > 0$ be smaller than the distance of all supports from zero. If $\varphi_\ell((-1)^j aW(\theta)/L) \neq 0$, then $a \geq 8cL$.

First cut at $a \leq L^D$, where $D > 2$. By (15), the total contribution of r-tuples for which one digit exceeds L^D and all the others exceed $8cL$ is

$$O\left(L^r \frac{1}{L^D L^{r-1}}\right) = O(L^{1-D}).$$

For $a \leq L^D$, the C^1 function $f_{a,j}(\theta) = \varphi_\ell((-1)^j aW(\theta)/L)$ satisfies $\|f'_{a,j}\|_\infty \leq C_\varphi a/L \leq C_\varphi L^{D-1}$. Jackson's theorem [11, Vol. I, Chap. III], with degree $M = L^{A_0+r+D+4}$, gives a trigonometric polynomial whose uniform error is $O(L^{-A_0-r-4})$. Its coefficient ℓ^1 norm is at most $(2M+1)(\|\varphi_\ell\|_\infty + 1)$ and hence is polynomial in L. Increasing the complexity exponent in Proposition 4.1, we may apply that proposition; after summation over $O(L^r)$ tuples, the approximation error is $O(L^{-A_0-4})$.

Non-good time tuples are $O_r(L^{r-1}H)$ in number. On each of them the probability that all factors are nonzero is $O_r(L^{-r})$ by (15); their total contribution is $O(H/L) = o(1)$. On good tuples, Proposition 4.1 factors the expectation, and its error remains $o(1)$ after summing $O(L^r)$ tuples. Equation (35) gives the vague convergence

$$L\mathbb{P}(AW(\Theta)/L \in dx) \Longrightarrow c_0 x^{-2}\, dx \quad (x > 0).$$

To evaluate (38), partition the ordered distinct tuples according to the unique permutation $\sigma \in S_r$ for which $j_{\sigma(1)} < \cdots < j_{\sigma(r)}$, and then according to the parity vector $\varepsilon_\ell = (-1)^{j_\ell} \in \{-1, 1\}$. For every fixed σ and $\varepsilon = (\varepsilon_1, \ldots, \varepsilon_r)$,

$$\#\left\{(j_1, \ldots, j_r) : j_{\sigma(1)} < \cdots < j_{\sigma(r)},\ (-1)^{j_\ell} = \varepsilon_\ell\right\} = \frac{1 + o(1)}{r!}\left(\frac{L}{2\lambda}\right)^r. \tag{39}$$

Indeed, the parity-restricted lattice discrepancy in the actual bulk interval is $O(L^{r-1})$, while replacing its length by L/λ contributes $O(HL^{r-1}) = o(L^r)$. Put

$$I_\ell^\varepsilon = \int_0^\infty \varphi_\ell(\varepsilon x) x^{-2}\, dx.$$

For a fixed ordering and parity vector, factorization and (35) show that the contribution tends to

$$\frac{1}{r!}\left(\frac{c_0}{2\lambda}\right)^r \prod_{\ell=1}^r I_\ell^{\varepsilon_\ell}.$$

Summing first over the $r!$ temporal orderings and then over all parity vectors yields

$$\prod_{\ell=1}^r \left\{\frac{c_0}{2\lambda}\int_{\mathbb{R}} \varphi_\ell(x)|x|^{-2}\, dx\right\} = \prod_{\ell=1}^r \int \varphi_\ell\, d\Lambda, \qquad \frac{c_0}{2\lambda} = \frac{1}{2\pi^2}.$$

Thus the ordered factorial moment measures converge to those of the Poisson random measure in (37). If $K \Subset \mathbb{R} \setminus \{0\}$ and $N_n(K)$ denotes the number of displayed points in K, the same iterated conditional tail bound gives, for every $r \geq 1$,

$$\sup_n \mathbb{E}(N_n(K))_r \leq C_K^r,$$

where $(x)_r = x(x-1)\cdots(x-r+1)$. In particular all local factorial moments are uniformly bounded, while the first moment supplies local tightness. Moreover, for $0 \leq u \leq 1$,

$$\sup_n \mathbb{E}(1+u)^{N_n(K)} = \sup_n \sum_{r \geq 0} \frac{u^r}{r!}\mathbb{E}(N_n(K))_r \leq \exp(C_K u).$$

Thus the local counts have uniformly bounded exponential moments. On finitely many disjoint compact continuity sets, the preceding calculation gives convergence of all mixed factorial moments to those of independent Poisson variables, and the exponential bound makes those moments distribution determining. Local tightness and an exhaustion by compact continuity sets now identify every subsequential limit as the Poisson random measure with intensity Λ; this is the standard factorial-moment criterion [6, Chap. 4]. □

We next control the compensated small jumps. Choose $\chi \in C^1([0, \infty))$ with $0 \leq \chi \leq 1$, $\chi = 1$ on $[0, 1]$, and $\chi = 0$ on $[2, \infty)$. For $0 < \varepsilon < 1$, set

$$U_{n,j}^{(\varepsilon)} = (-1)^j Z_{n,j}\chi\left(\frac{Z_{n,j}}{\varepsilon L}\right). \tag{40}$$

Lemma 5.2 (Small jumps)**.** *For a constant C independent of ε,*

$$\limsup_{n\to\infty} \frac{1}{L^2} \operatorname{Var}\left(\sum_{j \in \mathcal{J}_n} U_{n,j}^{(\varepsilon)}\right) \leq C\varepsilon. \tag{41}$$

Proof. Since $W \leq 1/8$, Lemma 3.1(ii) gives, uniformly in j, n,

$$\mathbb{P}(Z_{n,j} > t) \leq \frac{C}{1+t}.$$

For $j \neq k$, (15) similarly gives

$$\mathbb{P}(Z_{n,j} > s, Z_{n,k} > t) \leq \frac{C}{(1+s)(1+t)}.$$

Layer-cake integration, with $M = 2\varepsilon L$, yields

$$\mathbb{E}|U_{n,j}^{(\varepsilon)}|^2 \le C\varepsilon L, \qquad \mathbb{E}|U_{n,j}^{(\varepsilon)}U_{n,k}^{(\varepsilon)}| \le C\log^2(2+L). \tag{42}$$

The diagonal contribution is $O(\varepsilon L^2)$. Overlapping, H-near, or resonance-near pairs number $O(LH)$, and their total contribution is $O(LH\log^2 L) = o(L^2)$.

For the remaining pairs, cut each digit at $a \le A_L = L^D$. Since $|U_{n,j}^{(\varepsilon)}| \le 2\varepsilon L$,

$$\|U_{n,j}^{(\varepsilon)}\mathbf{1}_{\{a_{j+1}>A_L\}}\|_2 \le C\varepsilon L A_L^{-1/2}.$$

After centering, $\|X - \mathbb{E}X\|_2 \le 2\|X\|_2$; hence, after division by L, Minkowski's inequality shows that the sum of these tails is $O(\varepsilon L^{1-D/2}) = o(1)$ if $D > 2$. Fix an auxiliary exponent $A_0 > 0$, to be chosen sufficiently large. The map $z \mapsto z\chi(z/(\varepsilon L))$ has a Lipschitz constant bounded independently of ε and L on $[0,\infty)$, and W is 1/2-Lipschitz on $\mathbb{T}$. Hence, on $a \le A_L$, the truncated summand has Lipschitz norm at most $Ca \le CL^D$, uniformly in $0 < \varepsilon < 1$. Jackson approximation for Lipschitz functions, with uniform error L^{-A_0-4} and polynomial degree and coefficient norm, applies as in the proof of Proposition 5.1. Centering doubles the approximation error but does not change its order. Proposition 4.1, with A_0 chosen so large that $L^2 O_{A_0}(L^{-A_0}) = o(1)$, makes the sum of all remaining covariances $o(L^2)$. This proves (41). ☐

Corollary 5.3 (Principal Cauchy law)**.** *There are deterministic numbers $b_n^{(0)}$ such that*

$$\frac{1}{L}\sum_{j\in\mathcal{J}_n}(-1)^j Z_{n,j} - b_n^{(0)} \Longrightarrow \mathrm{Cauchy}\left(0, \frac{1}{2\pi}\right). \tag{43}$$

Proof. For fixed ε, split $x = x\chi(|x|/\varepsilon) + x(1-\chi(|x|/\varepsilon))$. Proposition 5.1 and the continuous mapping theorem give convergence of the second, large-jump part; the digit-tail bound gives the uniform finite-n estimate

$$\sup_n \mathbb{E}\Xi_n\{x : |x| > R\} \le C/R,$$

so the tails $|x| > R$ may be removed before applying continuous mapping. Center the small part by

$$b_{n,\varepsilon} = \frac{1}{L}\sum_{j\in\mathcal{J}_n}\mathbb{E}U_{n,j}^{(\varepsilon)}.$$

Put

$$R_{n,\varepsilon} = \frac{1}{L}\sum_{j\in\mathcal{J}_n}U_{n,j}^{(\varepsilon)} - b_{n,\varepsilon}, \qquad Y_{n,\varepsilon} = \int x\bigl(1-\chi(|x|/\varepsilon)\bigr)\,d\Xi_n(x).$$

Then the principal sum centered by $b_{n,\varepsilon}$ is exactly $Y_{n,\varepsilon} + R_{n,\varepsilon}$, and Lemma 5.2 and Chebyshev's inequality give, for every $\delta > 0$,

$$\limsup_{n\to\infty}\mathbb{P}(|R_{n,\varepsilon}| > \delta) \le \frac{C\varepsilon}{\delta^2}. \tag{44}$$

Thus the centered small part becomes negligible as $\varepsilon \downarrow 0$. The transition shell $\varepsilon < |x| < 2\varepsilon$ is included in the same continuous truncation; its limiting compensated variance is $O(\varepsilon)$.

For fixed ε, Proposition 5.1 and the preceding tail removal give $Y_{n,\varepsilon} \Longrightarrow Y_\varepsilon$. For the smooth cutoff used here, the characteristic exponent of Y_ε is, first on $|x| \le R$,

$$\int_{|x|\le R}(\exp\{itx(1-\chi(|x|/\varepsilon))\} - 1)\,d\Lambda(x).$$

Letting $R \to \infty$ is legitimate because the expected number of points outside $[-R,R]$ is $O(R^{-1})$. Symmetry cancels the imaginary part, and then dominated convergence as $\varepsilon \downarrow 0$ gives

$$\int_{\mathbb{R}}(\cos(tx) - 1)\frac{dx}{2\pi^2 x^2} = -\frac{|t|}{2\pi}.$$

Hence $Y_\varepsilon \Longrightarrow \mathrm{Cauchy}(0, 1/(2\pi))$ as $\varepsilon \downarrow 0$. The convergence-together criterion applied to this limit and (44) now proves the claim. Explicitly, choose $\varepsilon_m \downarrow 0$ so that $\varepsilon_m m^2 \to 0$ and the

Prokhorov distance from the law of Y_{ε_m} to the target law is at most $1/m$. Then choose $N_m \uparrow \infty$ so that, for $n \geq N_m$, the Prokhorov distance between the laws of Y_{n,ε_m} and Y_{ε_m} is at most $1/m$ and $\mathbb{P}(|R_{n,\varepsilon_m}| > 1/m) \leq 2C\varepsilon_m m^2$. Set $\varepsilon_n = \varepsilon_m$ for $N_m \leq n < N_{m+1}$, and define it arbitrarily for $n < N_1$. With $b_n^{(0)} := b_{n,\varepsilon_n}$ this gives (43). □

6. The carry graph and the bounded remainder

Let $(\widehat{\Omega}, \widehat{\nu}, \sigma)$ be the two-sided natural extension of the Gauss system. Concretely, up to null boundary sets, we use

$$\widehat{\Omega} = (0,1)^2, \qquad d\widehat{\nu}(x,y) = \frac{dx\,dy}{\log 2\,(1+xy)^2}, \qquad \sigma(x,y) = \left(Tx, \frac{1}{a_1(x)+y}\right),$$

and write $x(\omega) = [0; a_1(\omega), a_2(\omega), \ldots]$. Whenever order, ceiling, or fractional part is used for a torus coordinate, we identify $\mathbb{T}$ with its canonical representatives in $[0,1)$. Define the torus cocycle

$$\mathcal{S}(\omega, r, s) = (\sigma\omega, s, r - a_1(\omega)s \pmod 1). \tag{45}$$

The fibre matrix

$$A_a = \begin{pmatrix} 0 & 1 \\ 1 & -a \end{pmatrix}$$

lies in $\mathrm{GL}_2(\mathbb{Z})$ and preserves Haar measure on $\mathbb{T}^2$. The displayed density $\widehat{\nu}$ is σ-invariant by branchwise change of variables; hence $\widehat{\mu}_0 = \widehat{\nu} \otimes m_{\mathbb{T}^2}$ is invariant.

Lemma 6.1 (Endpoint-controlled continuant recurrence). *Let $a_i \geq 1$ and let an integer sequence with $(z_0, z_1) \neq (0,0)$ satisfy*

$$z_{i+2} = a_{i+1} z_{i+1} + z_i \qquad (0 \leq i < m).$$

If

$$\max(|z_0|, |z_1|, |z_m|, |z_{m+1}|) \leq K,$$

then $m \leq C \log(2K)$ for an absolute constant C.

Proof. Put $S_i = |z_i| + |z_{i+1}|$ for $0 \leq i \leq m$, and call i aligned if $z_i z_{i+1} \geq 0$. Since the recurrence is invertible, no adjacent pair is $(0,0)$, and hence $S_i \geq 1$. Alignment propagates forwards, because

$$z_{i+1} z_{i+2} = a_{i+1} z_{i+1}^2 + z_i z_{i+1} \geq 0.$$

Applying the recurrence twice at an aligned index gives the first inequality below. If $i+1$ and $i+2$ are both unaligned, solving the same two steps backwards gives the second:

$$S_{i+2} \geq 2S_i, \qquad S_i \geq 2S_{i+2} \qquad (i+2 \leq m),$$

with the respective alignment hypotheses just stated.

Let t be the first aligned index in $\{0, \ldots, m\}$, with $t = m+1$ if there is none. Set $k_- = \lfloor (t-1)/2 \rfloor$ for $t \geq 1$ and $k_- = 0$ for $t = 0$; set $k_+ = \lfloor (m-t)/2 \rfloor$ for $t \leq m$ and $k_+ = 0$ otherwise. Iteration on the unaligned prefix and aligned suffix gives

$$m \leq 2k_- + 2k_+ + 3, \qquad 2^{k_-} \leq S_0, \qquad 2^{k_+} \leq S_m.$$

Here the possible final unpaired step is harmless because S_i is nondecreasing on the aligned suffix. Since $S_0, S_m \leq 2K$ and the nonzero integer initial pair forces $K \geq 1$, taking logarithms now gives

$$m \leq \frac{4\log(2K)}{\log 2} + 3 \leq \frac{7}{\log 2}\log(2K).$$

This proves the claim, for example with $C = 7/\log 2$. □

Lemma 6.2 (Gauss–torus mixing). *The system $(\widehat{\Omega} \times \mathbb{T}^2, \widehat{\mu}_0, \mathcal{S})$ is mixing.*

Proof. It is enough to use finite digit-cylinder functions times torus characters. For characters $k, \ell \in \mathbb{Z}^2$, the fibre integral in an m-step correlation vanishes unless

$$k + (A_{a_m} \cdots A_{a_1})^{\mathrm{T}} \ell = 0. \tag{46}$$

Indeed, reverse the matrix order forced by the transpose and, for $0 \le i \le m$, put

$$w_i = A^{\mathrm{T}}_{a_{m-i+1}} \cdots A^{\mathrm{T}}_{a_m} \ell = (\xi_i, \eta_i)^{\mathrm{T}}, \qquad w_0 = \ell.$$

Since $A_a^{\mathrm{T}} = A_a$, for $0 \le i < m$,

$$w_{i+1} = A^{\mathrm{T}}_{a_{m-i}} w_i = (\eta_i, \xi_i - a_{m-i}\eta_i)^{\mathrm{T}}.$$

Thus $\xi_{i+1} = \eta_i$; on defining $\xi_{m+1} := \eta_m$, we have

$$\xi_{i+2} = \xi_i - a_{m-i}\xi_{i+1} \qquad (0 \le i < m).$$

Consequently $z_i = (-1)^i \xi_i$ satisfies $z_{i+2} = a_{m-i} z_{i+1} + z_i$, which is the recurrence of Lemma 6.1 after relabelling the positive coefficients. Moreover, $w_m = (A_{a_m} \cdots A_{a_1})^{\mathrm{T}} \ell = -k$ under (46); hence (z_0, z_1) is, up to signs, the coordinate pair of ℓ, and (z_m, z_{m+1}) is the coordinate pair of k. Since the matrix product in (46) is invertible, $\ell = 0$ if and only if $k = 0$. In the nonzero case $(z_0, z_1) \ne (0, 0)$, and both endpoint pairs are bounded in terms of the fixed vectors k and ℓ, so Lemma 6.1 applies. Thus the resonance relation is impossible for all sufficiently large m. The zero modes reduce to mixing of the Gauss natural extension; this follows from Lemma 3.1(i), since the natural extension of a mixing probability-preserving endomorphism is mixing. Density of cylinder functions times characters in L^2 completes the proof. □

We need to transfer fixed windows of the actual curve process to this stationary system. The complete quotient is an infinite-future coordinate, so we record the required full-state form rather than silently treating it as a finite digit function.

For a fixed $R \ge 0$, give

$$\mathcal{X}_R = \mathbb{N}^{2R+1} \times [0,1]^{2R+1} \times \mathbb{T}^{2R+2} \tag{47}$$

the product topology, with $\mathbb{N}$ discrete, and define the actual window

$$\mathcal{W}^{(R)}_{n,j} = \big((a_{j+t+1})_{t=-R}^{R}, (x_{j+t})_{t=-R}^{R}, (\theta_{j+t})_{t=-R-1}^{R} \big) \in \mathcal{X}_R. \tag{48}$$

The stationary window is defined from the two-sided Gauss–torus system in the same way, and its law is denoted by μ_R. Both laws are supported on the corresponding orbit-consistent Borel subset of $\mathcal{X}_R$.

For $R' \ge R$, let $\pi_{R',R} : \mathcal{X}_{R'} \to \mathcal{X}_R$ be the coordinate projection that retains the digit and real coordinates with $-R \le t \le R$ and the torus coordinates with $-R-1 \le t \le R$. Then $(\pi_{R',R})_* \mu_{R'} = \mu_R$, and, whenever both actual windows are defined, $\pi_{R',R}(\mathcal{W}^{(R')}_{n,j}) = \mathcal{W}^{(R)}_{n,j}$.

Lemma 6.3 (Uniform full-state transfer)**.** *Let $\eta^{(R)}_{n,j}$ be the law of* (48) *for Lebesgue α, and let μ_R be the stationary window law. If $G : \mathcal{X}_R \to \mathbb{C}$ is bounded and Borel measurable, and its discontinuity set has μ_R-measure zero, then*

$$\sup_{j \in \mathcal{J}_n} \left| \int G \, d\eta^{(R)}_{n,j} - \int G \, d\mu_R \right| \longrightarrow 0. \tag{49}$$

Proof. First take a digit-cylinder amplitude supported on finitely many local words, times a torus character $\exp(2\pi i \sum_{t=-R-1}^{R} c_t \theta_{j+t})$, and let d be the largest initial digit depth on which the chosen test depends. For the raw digit block in (48), $d = j + R + 1$; after replacing complete quotients by a fixed number M of future digits below, one may have $d \le j + R + M$. Partition into complete depth-d cylinders over the finitely many supported local words. By (31), on each such cylinder the coefficients A_w, B_w, the integer $Q_j = A_w q_j - B_w q_{j-1}$, and the whole block amplitude are fixed. If $A_w = B_w = 0$, uniform Gauss mixing gives the stationary mean. Otherwise choose $\kappa, \delta > 0$ with $\kappa + 3\delta < 80\lambda$ and apply Lemma 3.3 with $\eta = \mathrm{e}^{-\kappa H}$. The bound there is uniform in the nonzero pair (A_w, B_w), so after summing over the finitely many words a complete cylinder union of mass $O(\mathrm{e}^{-\kappa H}+F_{j+1}^{-2})=O(\mathrm{e}^{-cH})$ is removed, and on its complement $|Q_j|\ge \mathrm{e}^{-\kappa H}q_j$.

Put $t=\lfloor (m_n+j)/2-40H\rfloor$. Because $j\in\mathcal{J}_n$ and $d-j=O_{R,M}(1)$ for each fixed cylinder test, one has $t>d$ for all sufficiently large n, uniformly in j. Retain the coarser complete depth-j cylinder union on which $q_j\ge \mathrm{e}^{\lambda j-\delta H}$ and, below every retained depth-d prefix, the complete depth-t descendants on which $q_t\le \mathrm{e}^{\lambda t+\delta H}$. The discarded mass is $O(\mathrm{e}^{-c_\delta L^{1/2}})$ by (20). On every retained descendant,

$$\log\frac{q_t^2}{n|Q_j|}\le -(80\lambda-\kappa-3\delta)H+O(1),$$

and hence $q_t^2\le \mathrm{e}^{-c'_{R,G}H}n|Q_j|$ for some $c'_{R,G}>0$ and all large n. Lemma 3.4, with prefix depth d, therefore makes every nonzero character $O(\mathrm{e}^{-c'_{R,G}H}+\mathrm{e}^{-c_\delta L^{1/2}})$. This proves the required one-block convergence uniformly in j for finite sums of cylinder–character tests with finitely supported local digit amplitudes.

General finite-window tests are reduced to this class by digit truncation. In particular, first restrict the finitely many digits required by (31) and by the digit-cylinder amplitude to $a\le K$; Lemma 3.1(ii) and a union bound make the omitted mass $O_R(1/K)$, uniformly for the actual and stationary window laws. Continuous functions of the complete quotients are then included by replacing, simultaneously for every $|t|\le R$, the coordinate $x_{j+t}=[0;a_{j+t+1},a_{j+t+2},\dots]$ by its first M future digits. The maximum coordinate error is $O_R(F_M^{-2})$; hence, for continuous G, the test-function error is at most $\omega_G(C_RF_M^{-2})$. Restrict every digit appearing in these finite continued fractions to $a\le K$ as well. The total discarded mass is $O_R(M/K)$, uniformly for both window laws. These truncated tests are covered by the preceding argument with $d\le j+R+M$. For fixed M,K only finitely many words remain; first let $n\to\infty$, then $K\to\infty$, and finally $M\to\infty$.

For $K\ge 1$, let $\mathcal{X}_{R,K}\subset\mathcal{X}_R$ be the set on which all digit coordinates are at most K; it is compact, and these sets exhaust $\mathcal{X}_R$. The unital self-conjugate algebra generated by finite digit-cylinder indicators, continuous functions of the real coordinates, and torus characters separates points on each $\mathcal{X}_{R,K}$. Stone–Weierstrass therefore makes it uniformly dense there; the same algebra is convergence determining and its real span is dense in $L^2(\mu_R)$. The family $\eta_{n,j}^{(R)}$ is uniformly tight: real and torus coordinates are compact, and Lemma 3.1(ii) controls the finitely many digit coordinates. If (49) failed, tightness would give a weakly convergent subsequence, say with limit η. Convergence on the determining algebra forces $\eta=\mu_R$, and the Portmanteau theorem applied to the bounded μ_R-almost-everywhere continuous function G then contradicts the assumed failure. $\square$

For $z=(x,r,s)$ and $d\in\mathbb{N}_0$, define

$$\phi_z(d)=\lceil x(d+r)-s\rceil. \tag{50}$$

Equation (10) shows that the actual carries satisfy $d_{j+1}=\phi_{(x_j,\theta_{j-1},\theta_j)}(d_j)$. If $e=d+r$, the updated real carry obeys

$$xe\le e'<xe+1. \tag{51}$$

Combining this with (6), all actual trajectories and all pullback trajectories started from zero remain below an absolute integer D.

Set

$$\mathcal{R}=\left\{(x,r,s):x<\frac{1}{4(D+1)},\quad \frac12<s<\frac34\right\}. \tag{52}$$

For $0\le d\le D$ and $z\in\mathcal{R}$,

$$-\frac34<x(d+r)-s<-\frac14,$$

so $\phi_z(d)=0$. The set $\mathcal{R}$ has positive $\widehat{\mu}_0$-measure. By Lemma 6.2, almost every two-sided orbit visits it infinitely often in the past. Start at time $-N$ with carry zero and iterate to time zero.

After the most recent reset the answer no longer depends on N, so it stabilizes almost surely to a measurable function d_* satisfying

$$d_*(\mathcal{S}z) = \phi_z(d_*(z)). \tag{53}$$

Thus the stationary carry extension is a measurable invariant graph and is isomorphic to the mixing base–torus system.

For $R \geq 1$, define $d_j^{(R)}$ by setting the carry equal to zero at time $j - R$ and iterating R times. Let

$$p_R = \widehat{\mu}_0 \left(\bigcap_{t=0}^{R-1} \mathcal{S}^{-t} \mathcal{R}^c \right).$$

Ergodicity and $\widehat{\mu}_0(\mathcal{R}) > 0$ imply $p_R \to 0$. Whenever the actual orbit has a reset during $[j-R, j)$, $d_j = d_j^{(R)}$. The no-reset indicator has stationary-null boundary, so Lemma 6.3 gives

$$\sup_{j \in \mathcal{J}_n} \mathbb{P}(d_j \neq d_j^{(R)}) \leq p_R + o_n(1). \tag{54}$$

We now prove the weak law for the remainder in (11).

Proposition 6.4 (Bounded-remainder weak law)**.**

$$\frac{1}{L} \sum_{j \in \mathcal{J}_n} (-1)^j (B_j - \mathbb{E}B_j) \longrightarrow 0 \quad \textit{in probability.} \tag{55}$$

Proof. Let $B_j^{(R)}$ be the bounded remainder obtained by replacing d_j with $d_j^{(R)}$. Choose a bounded Borel function $B^{(R)} : \mathcal{X}_R \to \mathbb{R}$ whose restriction to the orbit-consistent support represents this remainder, so that $B_j^{(R)} = B^{(R)}(\mathcal{W}_{n,j}^{(R)})$. For fixed R, this function is bounded and μ_R-almost-everywhere continuous: its discontinuities lie on countably many Gauss boundaries and finitely many ceiling or fractional-part hypersurfaces, all of stationary measure zero.

Density bridge. Fix R and $\varepsilon > 0$. The $L^2(\mu_R)$-density established after Lemma 6.3 first gives a finite sum

$$G(\mathbf{a}, \mathbf{x}, \boldsymbol{\vartheta}) = \sum_{\ell=1}^{N} D_\ell(\mathbf{a}) g_\ell(\mathbf{x}) \exp\left(2\pi i \sum_{t=-R-1}^{R} c_{\ell,t} \vartheta_t \right),$$

where each D_ℓ is a finite digit-cylinder function, each g_ℓ is continuous on $[0,1]^{2R+1}$, and each $c_{\ell,t}$ is an integer, such that

$$\|B^{(R)} - G\|_{L^2(\mu_R)} < \varepsilon/4.$$

The torus range here is the full range $-R - 1 \leq t \leq R$ from (48).

For an integer $M \geq 1$, put $R_M = R + M$ and lift both functions to $\mathcal{X}_{R_M}$ by $\pi_{R_M,R}$. Since $(\pi_{R_M,R})_* \mu_{R_M} = \mu_R$, the first L^2 error remains less than $\varepsilon/4$ after this lift. On an orbit-consistent window replace, for every $-R \leq t \leq R$,

$$x_{j+t} = [0; a_{j+t+1}, a_{j+t+2}, \ldots] \quad \text{by} \quad x_{j+t}^{[M]} = [0; a_{j+t+1}, \ldots, a_{j+t+M}].$$

The continued-fraction contraction estimate gives $\max_{|t| \leq R} |x_{j+t} - x_{j+t}^{[M]}| = O_R(F_M^{-2})$. Since only finitely many continuous factors g_ℓ occur, their uniform continuity permits M to be chosen so that the resulting function G_M satisfies

$$\|G \circ \pi_{R_M,R} - G_M\|_{L^2(\mu_{R_M})} < \varepsilon/4.$$

Now let $E_{M,K}$ be the event that every digit in the full radius-R_M monomial word

$$a_{j-R_M+1}, \ldots, a_{j+R_M}$$

is at most K. This truncates all digits introduced by the finite-future replacement and also the M left-padding digits needed to regard the result as a radius-R_M monomial. By Lemma 3.1(ii), stationarity, and a union bound,

$$\mu_{R_M}(E_{M,K}^c) = O_R(M/K).$$

Since G_M is bounded, this measure estimate gives the required L^2 estimate explicitly:

$$\|G_M - G_M \mathbf{1}_{E_{M,K}}\|^2_{L^2(\mu_{R_M})} \le \|G_M\|^2_\infty \mu_{R_M}(E^c_{M,K}) = O_{R,G}(M/K).$$

Choose K so large that the corresponding L^2 norm is less than $\varepsilon/4$. For fixed M, K, only finitely many full digit words remain. For each summand ℓ and each such word $\widetilde{w}$, identity (31), applied to the central subword $w = (a_{j-R+1}, \dots, a_{j+R})$, gives

$$\sum_{t=-R-1}^{R} c_{\ell,t}\theta_{j+t} \equiv A_{\ell,w}\theta_j + B_{\ell,w}\theta_{j-1} \pmod 1.$$

Thus $G_M \mathbf{1}_{E_{M,K}}$ is exactly a finite linear combination of the monomials (32) at radius $R_M = R + M$: its digit amplitude is supported on the finitely many words $\widetilde{w}$, and its central Fourier modes are $(r, s) = (B_{\ell,w}, A_{\ell,w})$. These modes have finite support because both the original torus-character list and the retained word list are finite. The radius grows with M because the approximation to x_{j+R} uses digits through a_{j+R+M}; the additional left coordinates in the symmetric radius-R_M word are harmless and were truncated above.

Combining the three errors proves that, for every $\varepsilon > 0$, there are M, K and a finite linear combination P of monomials at radius $R + M$ such that

$$\|B^{(R)} \circ \pi_{R+M,R} - P\|_{L^2(\mu_{R+M})} < \varepsilon.$$

Because $B^{(R)} \circ \pi_{R+M,R}$ is real-valued,

$$|B^{(R)} \circ \pi_{R+M,R} - \operatorname{Re} P| \le |B^{(R)} \circ \pi_{R+M,R} - P|$$

pointwise. Adjoining the conjugate monomials shows that $\operatorname{Re} P$ is again a finite linear combination of monomials and is real-valued. Taking a sequence of accuracies decreasing to zero and choosing the truncation lengths increasingly, we let M range over that increasing sequence. We thereby obtain real-valued $P_{R,M}$ at radii $R_M = R + M$ such that

$$\|B^{(R)} \circ \pi_{R_M,R} - P_{R,M}\|_{L^2(\mu_{R_M})} =: \delta_{R,M} \longrightarrow 0. \tag{56}$$

For each fixed R, M, apply Lemma 6.3 with window radius R_M to the bounded, μ_{R_M}-almost-everywhere continuous function $|B^{(R)} \circ \pi_{R_M,R} - P_{R,M}|^2$ to obtain

$$\sup_{j\in\mathcal{J}_n} \mathbb{E}|B_j^{(R)} - P_{R,M,j}|^2 = \delta^2_{R,M} + o_n(1). \tag{57}$$

For fixed R, M, every $P_{R,M}$ is, by the bridge, a finite linear combination of the monomials (32); its radius, digit-word support, and Fourier support are all fixed before $n \to \infty$. Moreover, its integral against μ_{R_M} is its integral against $\widehat{\mu}_0$ under the stationary window map. Hence Proposition 4.2 applies to every covariance term in $\sum_{j\in\mathcal{J}_n} (-1)^j (P_{R,M,j} - \mathbb{E}P_{R,M,j})$. Its variance is $o(L^2)$: the diagonal contributes $O(L)$, the $O(LH)$ exceptional pairs contribute $O(LH) = o(L^2)$, and all other pairs contribute

$$O_{R,M}\Big(L^2\big(\mathrm{e}^{-c_{\mathrm{ld}}L^{1/2}} + \mathrm{e}^{-c_{\mathrm{osc}}H} + \rho^{c_{\mathrm{mix}}H}\big)\Big) = o(1).$$

Lemma 6.3 gives $\sup_j |\mathbb{E}P_{R,M,j} - \int P_{R,M}\, d\mu_{R_M}| = o(1)$, so replacing the stationary means in Proposition 4.2 by the actual means adds only $o(L^2)$ to the covariance sum.

Write $\widetilde{X} = X - \mathbb{E}X$. Minkowski's inequality and (57) give

$$\begin{aligned} &\left\| \frac{1}{L} \sum_{j\in\mathcal{J}_n} (-1)^j (\widetilde{B}_j^{(R)} - \widetilde{P}_{R,M,j}) \right\|_2 \\ &\qquad \le \frac{2}{L} \sum_{j\in\mathcal{J}_n} \|B_j^{(R)} - P_{R,M,j}\|_2 \le C\delta_{R,M} + o_n(1). \end{aligned} \tag{58}$$

First let $n \to \infty$, then $M \to \infty$. This proves the analogue of (55) with $B_j^{(R)}$ for every fixed R.

Finally, boundedness and (54) imply

$$\sup_{j\in\mathcal{J}_n} \|B_j - B_j^{(R)}\|_2 \le C\sqrt{p_R + o_n(1)}.$$

A second centered Minkowski estimate yields

$$\limsup_{n\to\infty} \left\| \frac{1}{L} \sum_{j\in\mathcal{J}_n} (-1)^j (\widetilde{B}_j - \widetilde{B}_j^{(R)}) \right\|_2 \le C\sqrt{p_R}. \tag{59}$$

Letting $R \to \infty$ completes the proof. The order of limits is $n \to \infty$, then $M \to \infty$, then $R \to \infty$. □

7. Stopping, centering, and completion of the proof

The continuous height has the exact form

$$C_j = \frac{n}{q_j + q_{j-1}x_j}, \qquad N_j = C_j - E_j, \qquad 0 \le E_j \le E_*. \tag{60}$$

Lemma 7.1 (Stopping time and end terms)**.** *With $H = L^{3/4}$,*

$$\tau_n = \frac{L}{\lambda} + O_{\mathbb{P}}(H). \tag{61}$$

Moreover,

$$\frac{1}{L}\left[S_n(\alpha) - \sum_{j\in\mathcal{J}_n} (-1)^j \Phi(x_j, u_j) \right] \longrightarrow 0 \quad \textit{in probability.} \tag{62}$$

Proof. Since $q_j \le q_j + q_{j-1}x_j < 2q_j$, (60) gives a deterministic bracketing. If $q_j \le n/(2(E_*+1))$, then $C_j \ge E_* + 1$ and hence $N_j \ge 1$. If $q_j > n$, then $C_j < 1$ and, because N_j is a nonnegative integer, $N_j = 0$. Thus τ_n lies between the hitting times at which q_j crosses these two fixed multiples of n. Applying (16) at $L/\lambda \pm CH$ to both thresholds, and then choosing C large, proves (61).

Also,

$$|\Phi(x,u)| \le \frac{1}{8x} + \frac{1}{2} \le \frac{a_1(x)}{8} + \frac{5}{8}. \tag{63}$$

Lemma 3.1(ii) therefore gives

$$\sup_{j,n} \mathbb{P}(|\Phi(x_j, u_j)| > t) \le \frac{C}{1+t}. \tag{64}$$

Let $I_n = \{0, \dots, \tau_n - 1\}$. On the event $|\tau_n - m_n| \le CH$, the symmetric difference $I_n \triangle \mathcal{J}_n$ is contained in the deterministic union

$$\{0 \le j < 200H + 1\} \ \cup\ \{m_n - (C+200)H - 2 \le j \le m_n + CH + 2\},$$

which has $O_C(H)$ positions. Moreover,

$$\left| S_n(\alpha) - \sum_{j\in\mathcal{J}_n} (-1)^j \Phi(x_j, u_j) \right| \le \sum_{j\in I_n \triangle \mathcal{J}_n} |\Phi(x_j, u_j)|,$$

so the same tail argument applies in both directions of the symmetric difference. The probability that one of these costs exceeds εL is $O_C(H/L) = o(1)$. By integration of (64), the expected sum of the costs truncated at εL is $O_C(H \log L)$; Markov's inequality after division by L makes this $o(1)$. Finally let C grow and use (61). □

Proof of Theorem 1.1. By Proposition 2.2, Corollary 5.3, and Proposition 6.4, there are deterministic c_n such that

$$\frac{1}{L}\sum_{j\in\mathcal{J}_n}(-1)^j\Phi(x_j,u_j)-c_n \Longrightarrow \mathrm{Cauchy}\left(0,\frac{1}{2\pi}\right). \tag{65}$$

Lemma 7.1 and Slutsky's theorem replace the bulk sum by S_n/L.

It remains to remove the deterministic centering. For almost every α,

$$\{k(1-\alpha)\}=1-\{k\alpha\}, \qquad S_n(1-\alpha)=-S_n(\alpha).$$

Thus $X_n = S_n/\log n$ has an exactly symmetric law. The family $X_n - c_n$ is tight by (65); symmetry makes $X_n + c_n$ tight as well. Since their deterministic center difference is $2c_n$, the sequence (c_n) is bounded. If $c_{n_k} \to c$, then $X_{n_k} \Rightarrow Z + c$, where $Z \sim \mathrm{Cauchy}(0, 1/(2\pi))$. The limit must be symmetric about zero, and a nondegenerate translated Cauchy law has this symmetry only when $c = 0$. Hence every convergent subsequence of (c_n) tends to zero, so $c_n \to 0$.

The Cauchy distribution is continuous; therefore convergence in distribution is equivalent to the asserted convergence of distribution functions. ☐

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AI-use disclosure

The author selected the problem and directed the iterative research and prompting process. The initial proposed answer, the proof architecture, and the substantive mathematical steps were generated by OpenAI GPT-5.6 Sol Pro and developed through iterative, human-directed AI collaboration. OpenAI Codex, under the author's direction, was used to assemble, edit, compile, and perform automated consistency checks on this manuscript. The author takes responsibility for the submitted manuscript.

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